Forecasting Hospital Inpatient Bed Occupancy

- Texas, California, Florida

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Why occupancy forecasting matters?

- Improve capacity planning
  Helps hospitals anticipate future bed demand and maintain appropriate capacity levels.

- Prepare for seasonal demand surges
  Enables hospitals to plan ahead for predictable increases in patient volume.

- Support operational decision-making
  Provides data-driven insights for planning and resource management.

Forecasting Inpatient Bed Occupancy

- **Objective:**
  Forecast weekly inpatient bed occupancy across U.S. regions.

- **Scope:**
  Focus on top three regions with highest average Inpatient Bed occupancy (TX, CA, FL).

- **Forecast Horizon:**
  3 months (~13 weeks)

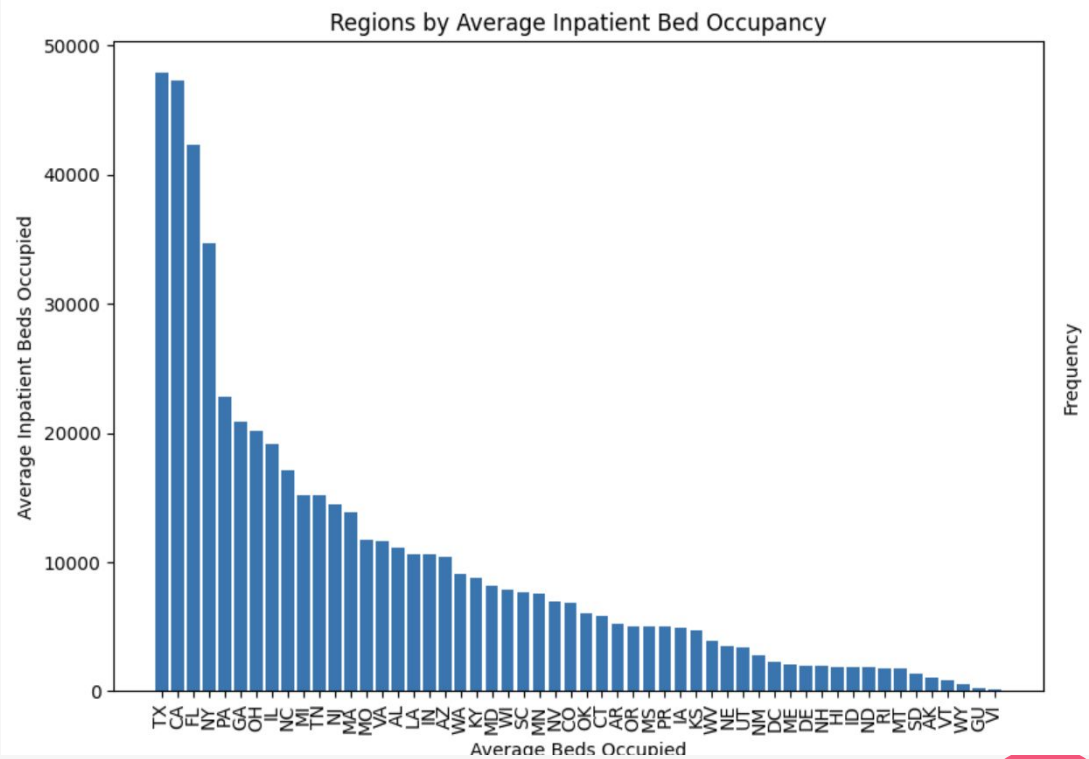
- **Outcome:**
  Identify forecasting models that provide reliable predictions for healthcare capacity planning.

Data Description

- Data Source: Kaggle: Weekly Hospital Respiratory Data and Metrics
  - Weekly hospital respiratory data and metrics aggregated at the national and state/territory levels, reported to the CDC's National Healthcare Safety Network (NHSN))
- Features:
  - 12,768 rows and 157 columns
  - Data across 56 U.S. regions (states, territories and federal district)
  - Reporting Periods (224 weeks):
    - Mandatory: Aug 2020 - Apr 2024;
    - Voluntary: May, 2024 - Oct, 2024
  - Admissions for COVID-19, Influenza, and RSV
  - Hospital Capacity and Occupancy
  - Reporting Coverage
  - Target Variable: Number of Inpatient Beds Occupied
  - Key Variables: Weekly Ending Date, Geographic aggregation, Number of Inpatient Beds, COVID-19 Admissions, Influenza Admissions, Reporting fields

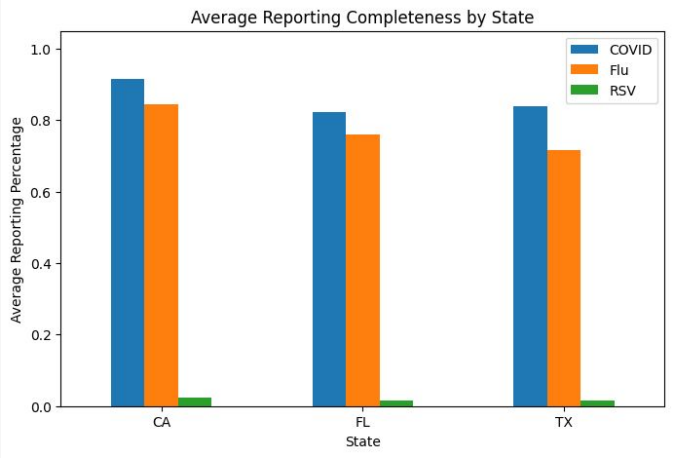
Regions by Inpatient Bed Occupancy

- Heavily right-skewed, with most observations between 10,000–20,000 beds
- **TX, CA, and FL** have the highest average occupancy
- Indicates substantial regional variation in hospital demand



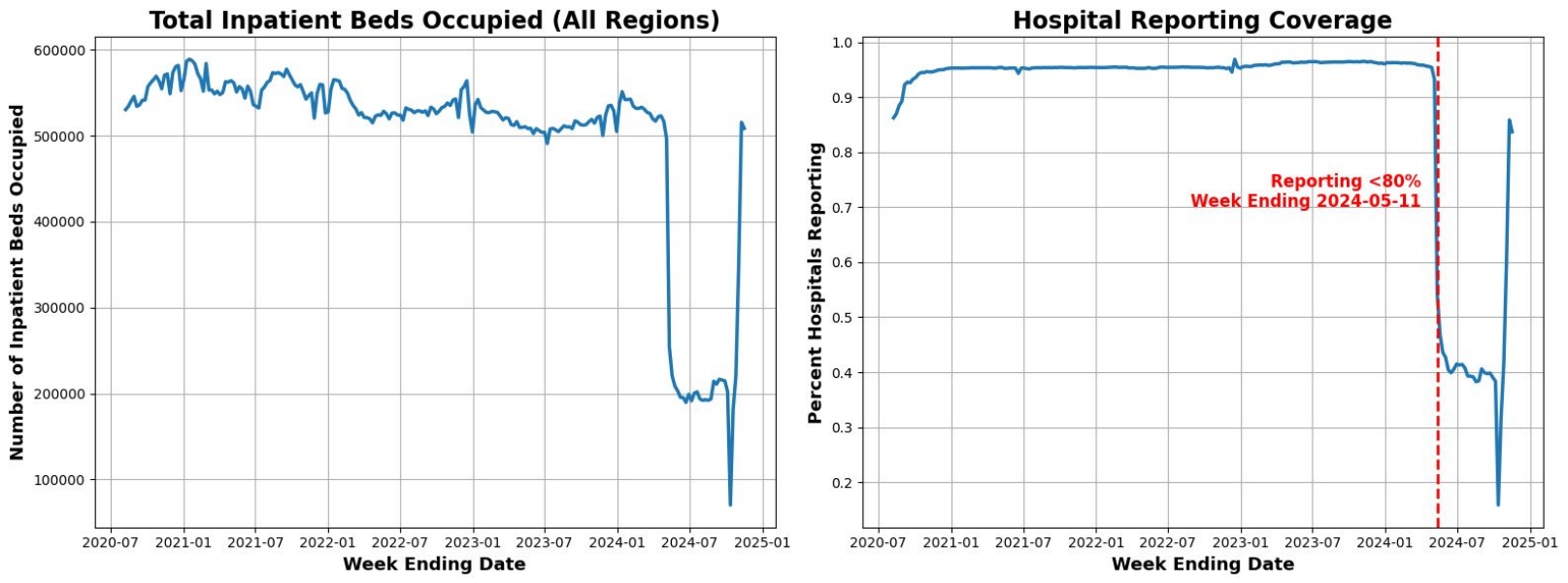
Data Preparation

- Removed non-relevant columns, retaining only key features and the target variable
- Standardized percentage reporting columns to a 0–1 scale (e.g., 92 → 0.92)
- Assessed reporting completeness for COVID, Flu, and RSV → removed RSV due to low reporting coverage
- Handles missing values by using time-series linear interpolation, followed by forward and backward filling



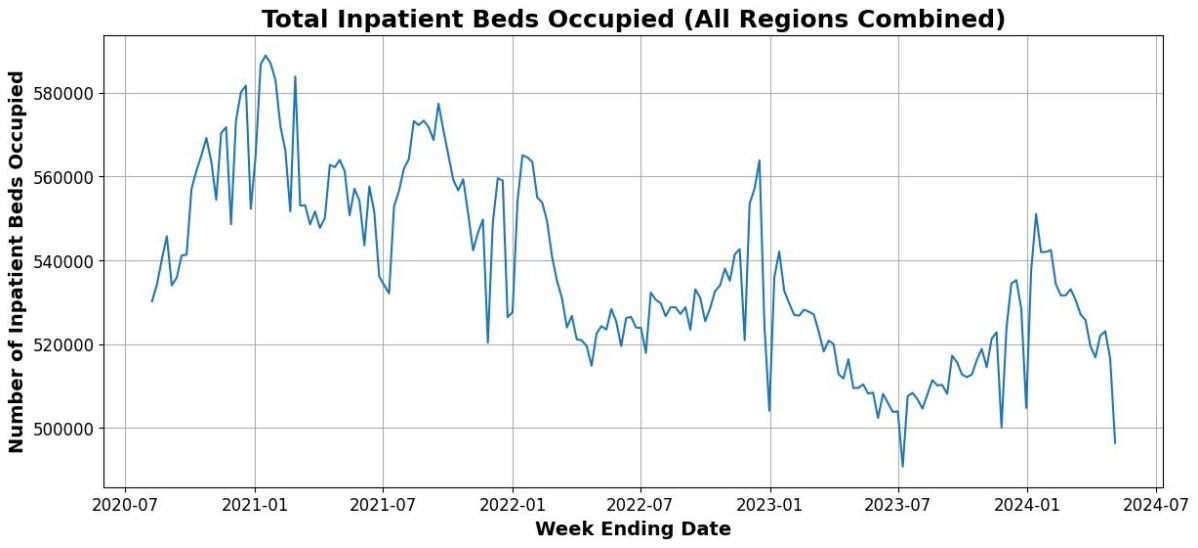
	Missing Values	Percent Missing
Total Patients Hospitalized with Influenza	509	4.14
Total Influenza Admissions	501	4.07
Total Patients Hospitalized with COVID-19	13	0.11
Total COVID-19 Admissions	13	0.11

Data Challenges – Reporting Policy Change



- Reporting coverage drops below **80% starting May 11, 2024** when reporting becomes **voluntary (1 May – 31 Oct 2024)**
- The apparent decline in inpatient beds reflects **reduced reporting rather than a true drop in occupancy**, so this period was removed from the analysis.

Validated Time Series Data (All Regions)



- **Cleaned and validated dataset** after addressing anomalies and reporting gaps
- **Clear COVID patterns:** higher occupancy during the early pandemic period
- **Trend stabilization:** gradual decline and stabilization in later periods

Modeling Approach

Model Development

Forecast Setup:
Horizon: 13 weeks (May 11 – Aug 3, 2024)
Train–Test Split: Last 13 weeks used as test set

Time Series Diagnostics:
Stationarity Test: Augmented Dickey-Fuller (ADF)
Correlation: ACF and PACF plots

Baseline Model:
Future value = last observed value

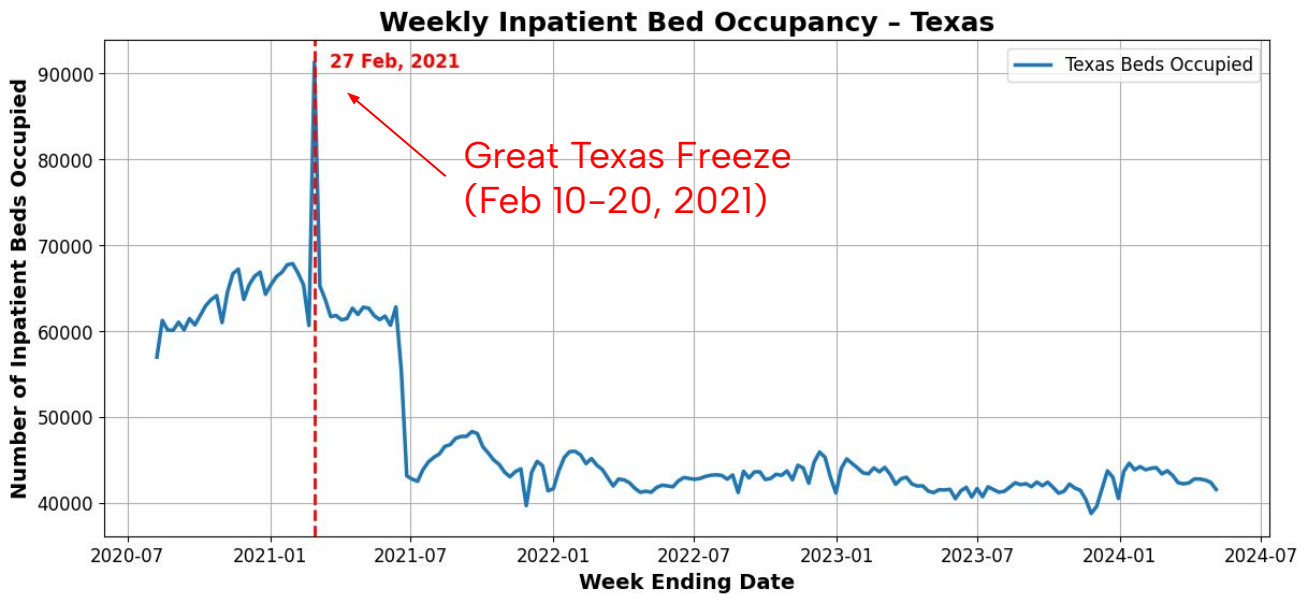
Forecasting Models:
ARIMA (p,d,q); SARIMA; ETS / Seasonal ETS

Model Evaluation & Forecasting

Model Evaluation:
Information Criteria: AIC, BIC
Forecast Accuracy: RMSE, MAPE
Residual Diagnostics: ACF of residuals

Final Forecast:
Best-performing model selected
13-week forecast generated

TX: Weekly Inpatient Bed Occupancy



- High healthcare demand during COVID-19 pandemic.
- The time series suggests a structural shift from pandemic-driven high utilization to a stable post-pandemic baseline.

TX: Stationarity & Model Identification

Hypothesis:

Ho (Null): Time series has a unit root (non-stationary, trend)

Ha (Alternative): Time series has no unit root (stationary, stable)

Initial ADF test Results (d=0):

ADF Statistic: -1.47

p-value: 0.55

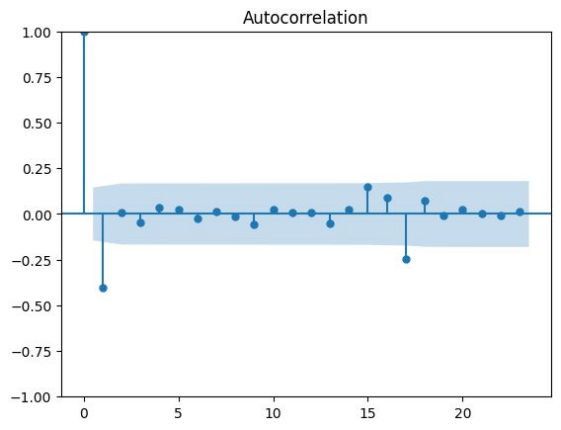
→ Fail to reject Ho. Series is nonstationary

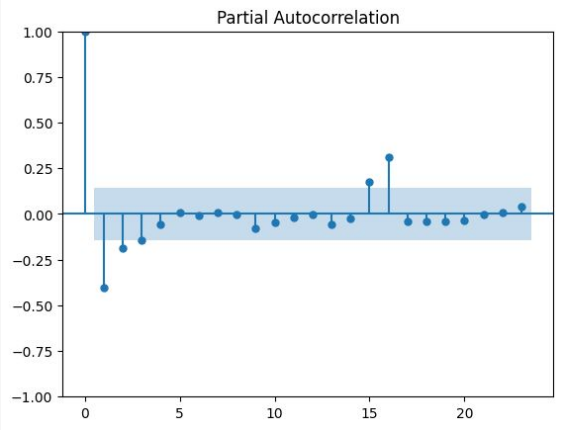
After Differencing (d=1):

ADF Statistic: -10.89

p-value: 1.29e-19

→ Reject Ho. Series is stationary





TX: ARIMA Model Comparison (AIC/BIC)

- **Compared multiple ARIMA models using AIC and BIC**, where lower values indicate a better model fit.
- **ARIMA(1,1,0) had the lowest AIC and BIC**, suggesting the best balance between accuracy and model complexity.

Model	AIC	BIC
ARIMA(0,1,1)	3475.48	3481.88
ARIMA(1,1,0)	3475.11	3481.52
ARIMA(1,1,1)	3476.91	3486.52
ARIMA(2,1,0)	3477.00	3486.61

TX: Model Comparison (test set)

Model	RMSE	MAPE
Baseline	1214.61	2.44
ARIMA *(1,1,0)*	1248.90	2.51
ETS	531.08	0.99
Seasonal ETS	2820.14	3.26
SARIMA	1193.69	2.40
SARIMAX *(2,1,1)(1, 0, 0, 52)*	945.54	1.86

Key Insights:

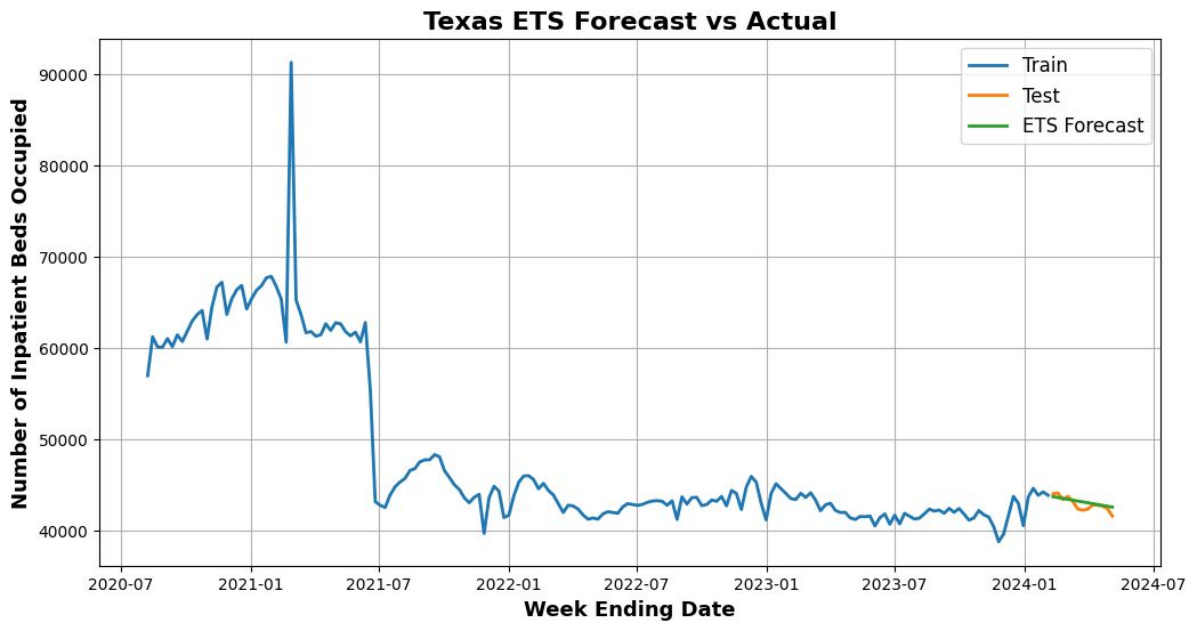
Forecast models evaluated using **RMSE and MAPE** on the test dataset.

ETS significantly outperforms other models, with the lowest prediction error

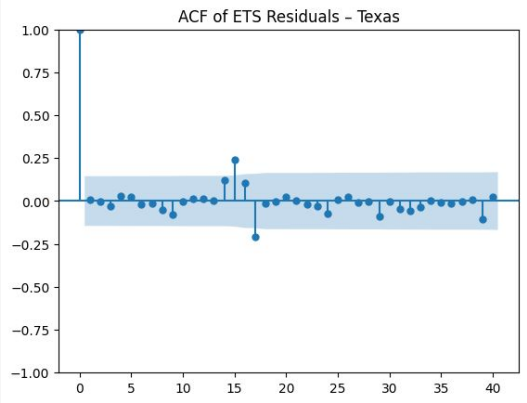
Suggests Texas bed occupancy follows a **stable level pattern suited for exponential smoothing.**

TX: ETS Forecast Vs Actual

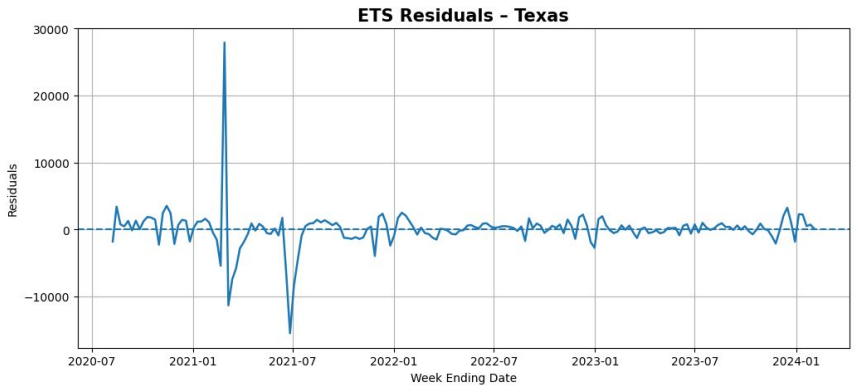
- ETS captures the **stable post–2022** occupancy level in Texas
- Forecast closely **follows the recent trend in the test period**
- **Low forecasting error (RMSE: 531, MAPE: 0.99%)** confirms strong performance



TX: Residual Diagnostics – ETS Model



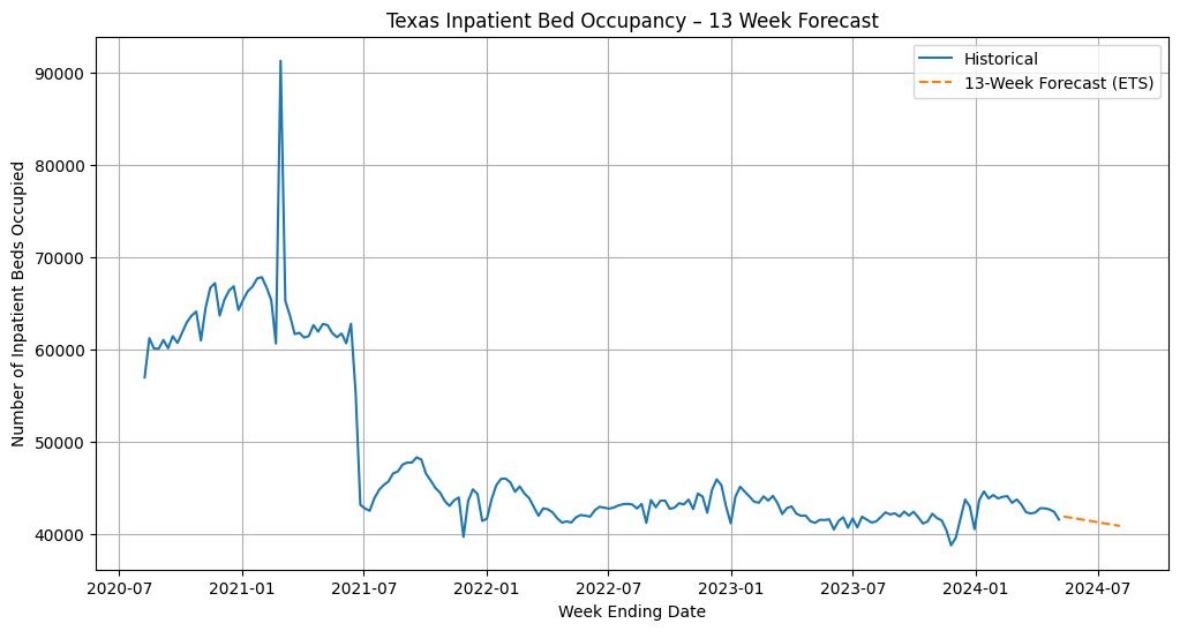
- Residual autocorrelations mostly within confidence interval
- No strong remaining temporal dependence
- Model assumptions satisfied



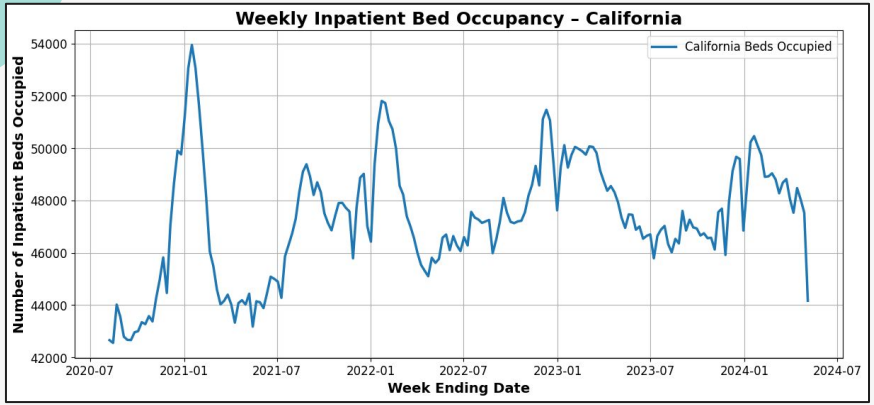
- Residuals fluctuate around zero
- A few large spikes appear early in the series
- Residuals become stable afterward

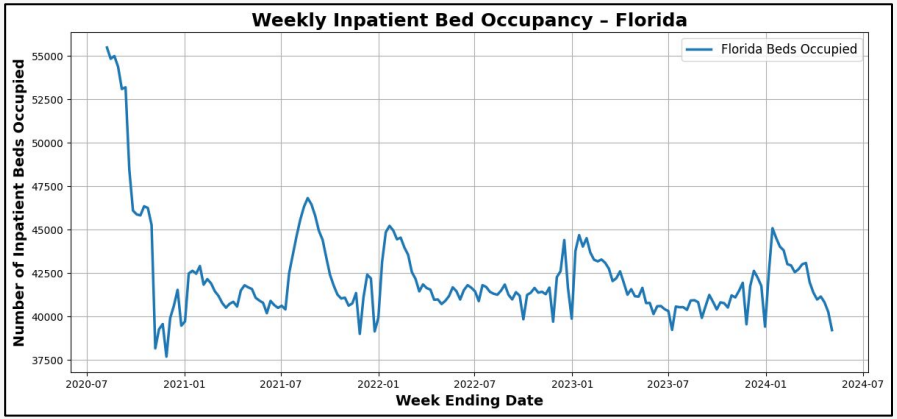
TX: 13-Week Forecast

- ETS model used to generate **13-week out-of-sample forecast**
- Forecast indicates **stable hospital utilization levels in Texas**
- Suggests **short-term demand remains predictable without major spikes**



CA & FL: Inpatient Bed Occupancy





- **California:** Relatively stable occupancy with moderate fluctuations over time.
- **Florida:** Larger structural drop early in the period followed by a stable range afterward.
- Differences in trend and variability help guide the **forecasting model selection for each region**.

CA & FL: Model Comparison

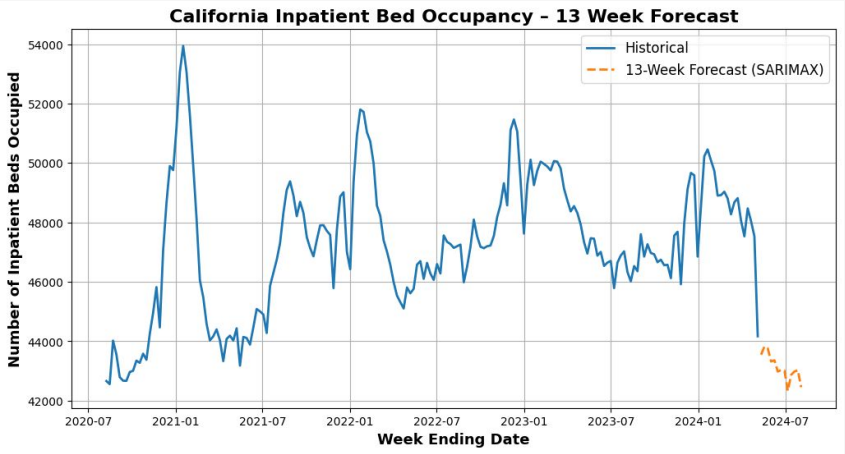
Model	CA		FL	
	RMSE	MAPE	RMSE	MAPE
Baseline	2054.81	3.49	2365.75	4.97
ARIMA	1067.03*(2,0,0)*	1.2	1882.7 *(1,0,1)*	3.87
ETS	2337.81	4.06	1820.29	3.77
Seasonal ETS	3749.95	7.11	1004.26	1.91
SARIMA	1267.88	1.97	1665.71	3.40
SARIMAX *(2,0,0)(1,0,0,52)*	877.99	1.15	1921.98	3.90

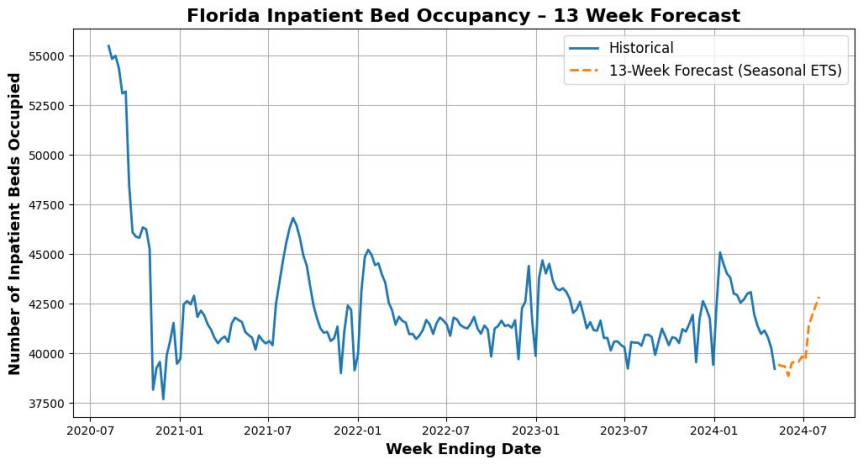
California: SARIMAX performed best, capturing **seasonality and external demand drivers**.

Florida: Seasonal ETS performed best, indicating **strong seasonal patterns**

CA & FL: 13-Week Forecast

- **California (SARIMAX):** Forecast suggests a **slight decline and stabilization** in inpatient bed occupancy over the next 13 weeks.
- **Florida (Seasonal ETS):** Forecast shows **moderate fluctuations with a small upward adjustment**, reflecting seasonal patterns.





Forecast Insights

TX

Forecast: Stable / Declining

- Forecast indicates relatively steady inpatient bed occupancy over the next 13 weeks.
- ETS performed best → series is primarily driven by a strong underlying trend.

CA

Forecast: Stabilizing

- Forecast shows a slight decline followed by stable occupancy levels.
- SARIMAX performed best → seasonality + external drivers

FL

Forecast: Moderate Increase

- Seasonal fluctuations expected
- Seasonal ETS performed best → recurring seasonal patterns strongly influence hospital occupancy.

Operational Implications by Region

Texas

- Forecast suggests **stable to slightly declining occupancy**, indicating current hospital capacity is likely sufficient in the short term.
- Hospitals can focus on **maintaining operational efficiency and optimizing resource allocation**.

California

- Forecast indicates **stabilizing occupancy levels**, suggesting relatively balanced hospital demand.
- Healthcare systems can focus on **maintaining steady staffing and resource distribution**.

Florida

- Forecast suggests a **moderate increase in occupancy**, reflecting seasonal demand patterns.
- Hospitals may need to **prepare additional staffing and resources to accommodate potential demand increases**.

Industry Applications

Public Health & Emergency Preparedness

Surges to moster likely hospital.

Planning to focus on maintaining **operational** efficiency.

Example:
Hospitals can focus u with public health & emergency preparedness.

Healthcare Resource Optimization

Optimize the **stabiliated** hospital occupancy.

on maintaining steady staffing and resource efficiency.

Example:
Hospitals can focus on health care resource optimization.

Data-Driven Policy Decisions

Supply chain to modicate pill tion of nentral demand.

Supply chain to policy consistrence and decisions.

Example:
Hospitals may need to avoid data-data-driven policy statements.

Industry Applications

Public Health & Emergency Preparedness

- **Anticipate surges** from respiratory illnesses such as COVID-19 and influenza
- Support **healthcare planning and emergency response**
- Enable **early intervention strategies** during outbreaks

Example: health departments can activate emergency capacity earlier.

Healthcare Resource Optimization

- **Optimize ICU beds**, ventilators, and medical equipment allocation
- Improve supply chain **planning for medications and PPE**
- **Reduce hospital wait times** and patient overflow

Example: hospitals can prepare ICU resources weeks in advance.

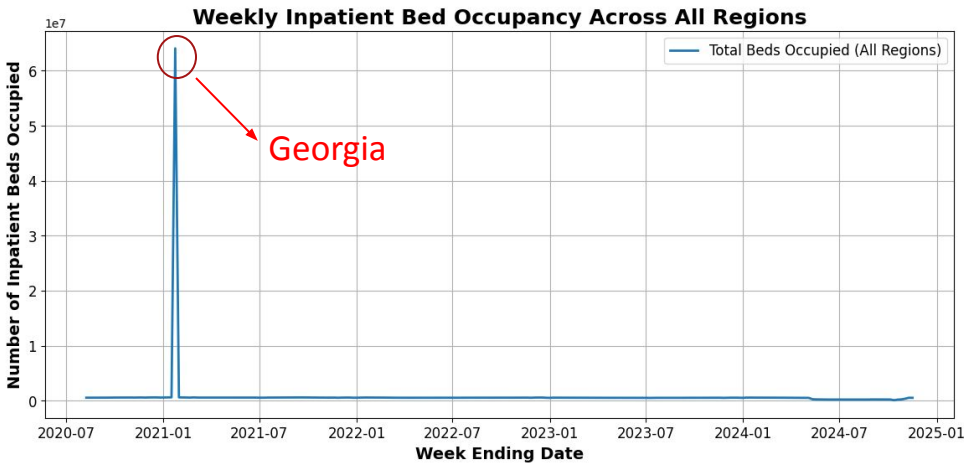
Data-Driven Policy Decisions

- Support state and federal **healthcare policy planning**
- Identify regions with higher healthcare demand volatility
- Improve **long-term healthcare infrastructure planning**

Example: policymakers can allocate funding to high-demand regions.

Data Challenge – Outlier in Georgia

- A sharp spike appears in the **weekly inpatient bed occupancy time series**.
- Investigation shows the anomaly originates from **Georgia (GA)**.
- Value exceeds total available beds and deviates from surrounding weeks.
- Replaced using linear interpolation.



Week Ending Date	Number of Inpatient Beds Occupied	Number of Inpatient Beds
2021-01-16	26283.0	31421.0
2021-01-23	63483001.0	31470.0
2021-01-30	25921.0	31671.0